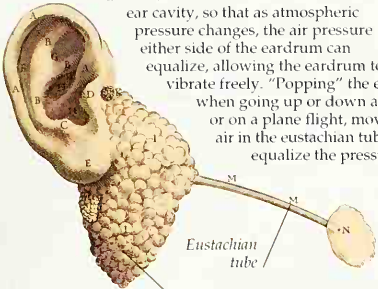
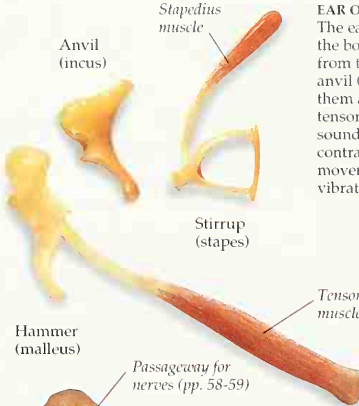
WHY EARS "POP"

Valsalva named the eustachian tube in this drawing from his *On the Human Ear*. The tube allows air from the throat into the middle ear cavity, so that as atmospheric pressure changes, the air pressure either side of the eardrum can equalize, allowing the eardrum to vibrate freely. "Popping" the ears when going up or down a hill, or on a plane flight, moves air in the eustachian tube to equalize the pressure.





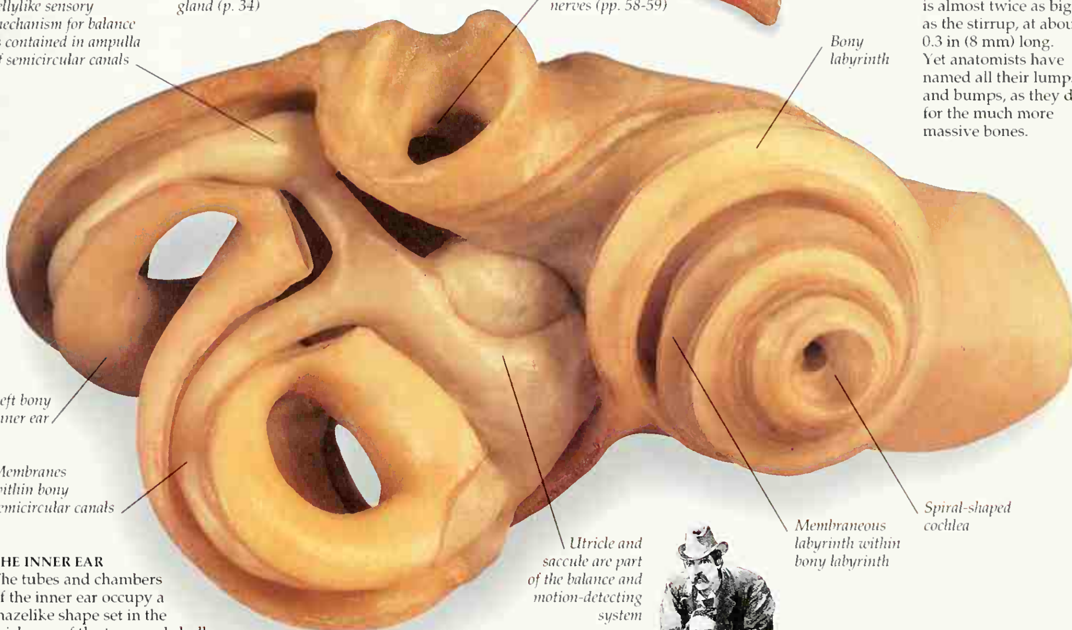
EAR OSSICLES

The ear ossicles spanning the middle ear are the body's smallest bones. They are named from their shapes as the hammer (malleus), anvil (incus), and stirrup (stapes). Attached to them are two of the body's smallest muscles, the tensor tympani and stapedius. If a very loud sound reaches the eardrum, these muscles contract. They damp down the eardrum's movements, and their own, to prevent over strong vibrations from damaging the delicate inner ear.

Hammer on a fingertip (actual size)

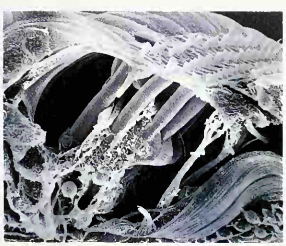
SMALLEST BONES

The ear ossicles are very tiny. The hammer is almost twice as big as the stirrup, at about 0.3 in (8 mm) long. Yet anatomists have named all their lumps and bumps, as they do for the much more massive bones.



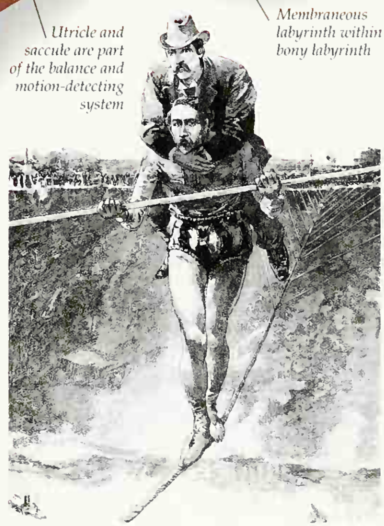
THE INNER EAR

The tubes and chambers of the inner ear occupy a mazelike shape set in the thickness of the temporal skull bone. This was named the osseous, or bony, labyrinth by Gabriele Fallopius. He also named the cochlea, from the Latin name for a snail's shell. The bony labyrinth is filled with fluid, perilymph. This surrounds a set of membranes, the membraneous labyrinth, which sits inside the bony labyrinth, following its shape. Inside the membraneous labyrinth is another fluid, endolymph.



ORGAN OF CORTI

Named after anatomist Alfonso Corti (1822-1888), this V-shaped set of membranes spirals inside the cochlea. More than 15,000 hair-bearing cells are in rows along its base membrane. Different areas of hair cells produce nerve signals when vibrated by high or low frequencies in the fluid around them, from high- or low-pitched sounds.



BALANCING ACT

The inner ear detects motion and balance as well. Head movements swirl fluid inside the semicircular canals. In these are microscopic blobs of jelly, in which are embedded hairs of sensory cells, which fire nerve signals to the brain. The canals are at right angles, so that each detects motion in one of the three dimensions of space. Patches of hair cells in the utricle and saccule are able to detect changes in motion and gravity.